BS-M301 — Probability and Statistics

Semester III • 3L:0T:3P • 3 Credits

Module 3 Exam Notes: Bivariate Distributions, Sums, Quotients and Bayes' Rule

CGEC | B.Tech ECE (2023–27) | *Biraj Sarkar*

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1. Bivariate Distributions and Their Fundamental Properties

In numerous engineering problems, physical phenomena involve the simultaneous observation of two interdependent random quantities, such as the in-phase and quadrature components of an RF communication signal (I, Q), transmitter voltage and received current (V, I), or temperature and humidity. These are modeled jointly as a **two-dimensional (bivariate) random variable** $(X, Y) : \Omega \to \mathbb{R}^2$.

Definition: Joint Probability Mass and Density Functions

Discrete Bivariate Case (Joint PMF): Let (X, Y) take values in a countable set of pairs (x_i, y_j). The **Joint PMF** is:

$$p_{XY}(x_i, y_j) = P(X = x_i, Y = y_j) \tag{1}$$

Axioms: (1) $p_{XY}(x_i, y_j) \geq 0, \quad \forall (i, j),$ (2) $\sum_i \sum_j p_{XY}(x_i, y_j) = 1.$

Continuous Bivariate Case (Joint PDF): (X, Y) is jointly continuous if there exists an integrable surface $f_{XY} : \mathbb{R}^2 \to \mathbb{R}$ such that for any two-dimensional region $D \subseteq \mathbb{R}^2$:

$$P((X, Y) \in D) = \iint_D f_{XY}(x, y)\, dxdy \tag{2}$$

Axioms:

$$\textbf{Non-negativity:} \quad f_{XY}(x, y) \geq 0, \quad \forall (x, y) \in \mathbb{R}^2 \tag{3}$$

$$\textbf{Unit Volume:} \quad \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{XY}(x, y)\, dxdy = 1 \tag{4}$$

1.1 Joint Cumulative Distribution Function (Joint CDF)

The **Joint CDF** of (X, Y) is defined as the probability of the infinite quadrant to the south-west of (x, y):

$$F_{XY}(x, y) = P(X \leq x, Y \leq y) = \begin{cases} \sum_{x_i \leq x} \sum_{y_j \leq y} p_{XY}(x_i, y_j), & \text{(Discrete)} \\ \int_{-\infty}^{x} \int_{-\infty}^{y} f_{XY}(u, v)\, dvdu, & \text{(Continuous)} \end{cases} \tag{5}$$

Properties of the Joint CDF and Recovery of Joint PDF

1. **Boundary Limits:**

$$F_{XY}(-\infty, y) = 0, \qquad F_{XY}(x, -\infty) = 0, \qquad F_{XY}(-\infty, -\infty) = 0 \tag{6}$$

$$F_{XY}(+\infty, +\infty) = 1 \tag{7}$$

2. **Marginal Recovery from Joint CDF:**

$$F_X(x) = \lim_{y \to +\infty} F_{XY}(x, y) = F_{XY}(x, \infty), \qquad F_Y(y) = \lim_{x \to +\infty} F_{XY}(x, y) = F_{XY}(\infty, y) \tag{8}$$

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3. **Recovery of Continuous Joint PDF by Partial Differentiation:** At every continuity point of $f_{XY}(x, y)$:

$$f_{XY}(x, y) = \frac{\partial^2}{\partial x \partial y} F_{XY}(x, y) = \frac{\partial^2}{\partial y \partial x} F_{XY}(x, y) \tag{9}$$

4. **Rectangle Probability Rule:** For any rectangular region $[x_1, x_2] \times [y_1, y_2]$:

$$P(x_1 < X \leq x_2,\ y_1 < Y \leq y_2) = F_{XY}(x_2, y_2) - F_{XY}(x_1, y_2) - F_{XY}(x_2, y_1) + F_{XY}(x_1, y_1) \geq 0 \tag{10}$$

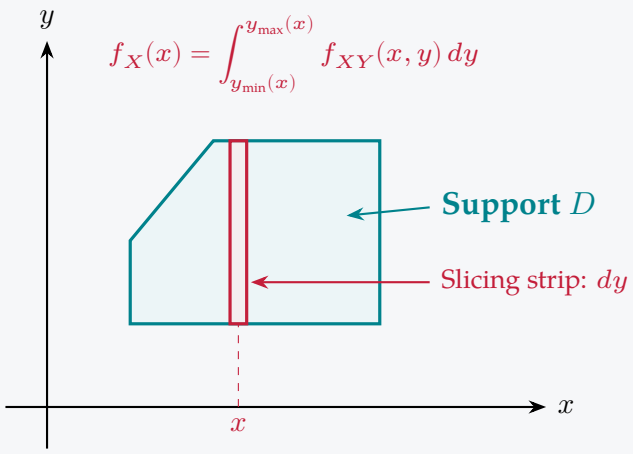
1.2 Marginal Distributions

The individual distribution of one variable obtained by integrating or summing out the other variable over its entire support is termed the **marginal distribution**.

$$\textbf{Marginal PMFs:} \quad p_X(x_i) = \sum_j p_{XY}(x_i, y_j), \qquad p_Y(y_j) = \sum_i p_{XY}(x_i, y_j) \tag{11}$$

$$\textbf{Marginal PDFs:} \quad f_X(x) = \int_{-\infty}^{\infty} f_{XY}(x, y)\, dy, \qquad f_Y(y) = \int_{-\infty}^{\infty} f_{XY}(x, y)\, dx \tag{12}$$

TikZ Schematic: Geometric Interpretation of Marginal Integration across Joint Support



Area under the cross-sectional slice at fixed x gives the marginal density $f_X(x)$.

2. Conditional Distributions, Densities and Bayes' Rule

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Definition: Conditional Probability Density Function

Given that the random variable Y has taken on a specific numerical value $Y = y$ (where marginal density $f_Y(y) > 0$), the **conditional PDF** of X given $Y = y$ is defined as:

$$f_{X|Y}(x \mid y) = \frac{f_{XY}(x,y)}{f_Y(y)} = \frac{f_{XY}(x,y)}{\displaystyle\int_{-\infty}^{\infty} f_{XY}(u,y)\,du} \tag{13}$$

Symmetrically, the conditional PDF of Y given $X = x$ (with $f_X(x) > 0$) is:

$$f_{Y|X}(y \mid x) = \frac{f_{XY}(x,y)}{f_X(x)} = \frac{f_{XY}(x,y)}{\displaystyle\int_{-\infty}^{\infty} f_{XY}(x,v)\,dv} \tag{14}$$

Every conditional density satisfies the standard PDF axioms: $f_{X|Y}(x \mid y) \geq 0$ and $\int_{-\infty}^{\infty} f_{X|Y}(x \mid y)dx = 1$.

2.1 Conditional Expectation and the Tower Property

- **Conditional Expectation:** The mean of X given $Y = y$ is a function of y:

$$E[X \mid Y = y] = \int_{-\infty}^{\infty} x f_{X|Y}(x \mid y)\,dx \tag{15}$$

- **Law of Iterated Expectations (Tower Property):**

$$E\left[E[X \mid Y]\right] = E[X] \tag{16}$$

Proof:

$$\begin{aligned} E\left[E[X \mid Y]\right] &= \int_{-\infty}^{\infty} \left(\int_{-\infty}^{\infty} x \frac{f_{XY}(x,y)}{f_Y(y)} dx \right) f_Y(y)\,dy \\ &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x f_{XY}(x,y)\,dxdy = \int_{-\infty}^{\infty} x \left(\int_{-\infty}^{\infty} f_{XY}(x,y)dy \right) dx \\ &= \int_{-\infty}^{\infty} x f_X(x)\,dx = E[X] \end{aligned} \tag{17}$$

- **Law of Total Variance (Eve's Law):**

$$\mathrm{Var}(X) = E[\mathrm{Var}(X \mid Y)] + \mathrm{Var}(E[X \mid Y]) \tag{18}$$

Total uncertainty equals the average unexplained variance plus the variance explained by Y.

2.2 Bayes' Rule for Continuous and Mixed Random Variables

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Continuous and Mixed Formulations of Bayes' Rule

Continuous Inversion (Posterior Density): Equating $f_{XY}(x,y) = f_X(x)f_{Y|X}(y \mid x) = f_Y(y)f_{X|Y}(x \mid y)$:

$$f_{X|Y}(x \mid y) = \frac{f_X(x) \cdot f_{Y|X}(y \mid x)}{f_Y(y)} = \frac{f_X(x) \cdot f_{Y|X}(y \mid x)}{\int_{-\infty}^{\infty} f_X(u) \cdot f_{Y|X}(y \mid u)\, du} \tag{19}$$

where $f_X(x)$ is the *prior density*, $f_{Y|X}(y \mid x)$ is the *likelihood function*, and $f_{X|Y}(x \mid y)$ is the *posterior density*.

Discrete Hypothesis with Continuous Measurement (Signal Detection): Let $\{H_1, H_2, \ldots, H_n\}$ be discrete transmitted symbols (priors $P(H_i)$) observed through continuous additive noise Y:

$$P(H_k \mid Y = y) = \frac{P(H_k) \cdot f_{Y|H_k}(y)}{\sum_{i=1}^{n} P(H_i) \cdot f_{Y|H_i}(y)} \tag{20}$$

3. Bivariate Transformations and the Jacobian Method

When analyzing signal processing systems, inputs (X, Y) are mapped into new output variables (U, V) via mathematical functions $U = g_1(X, Y)$ and $V = g_2(X, Y)$.

Theorem: Bivariate Jacobian Transformation

Let (X, Y) have joint PDF $f_{XY}(x, y)$ supported on set D. Let the transformation:

$$u = g_1(x, y), \qquad v = g_2(x, y) \tag{21}$$

define a one-to-one and invertible mapping from D onto range S, with unique inverse:

$$x = h_1(u, v), \qquad y = h_2(u, v) \tag{22}$$

If the partial derivatives are continuous, the **Jacobian determinant** J of the inverse transformation is:

$$J = \frac{\partial(x, y)}{\partial(u, v)} = \det \begin{pmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{pmatrix} = \frac{\partial x}{\partial u}\frac{\partial y}{\partial v} - \frac{\partial x}{\partial v}\frac{\partial y}{\partial u} \tag{23}$$

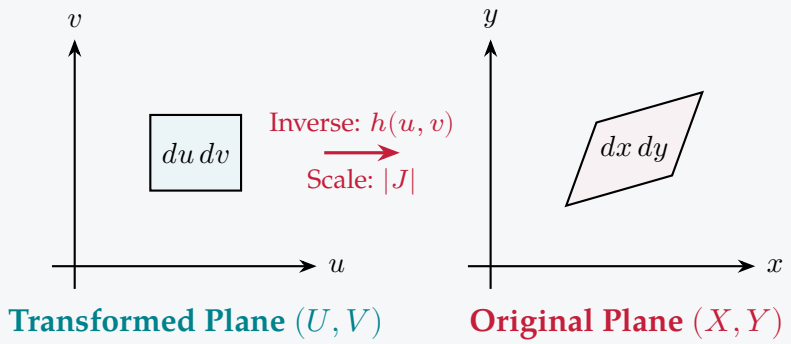
Then the joint PDF of (U, V) over S is given by:

$$f_{UV}(u, v) = f_{XY}\left(h_1(u, v), h_2(u, v)\right) \cdot |J| \tag{24}$$

where $|J|$ is the absolute value of the Jacobian determinant (representing the local area scaling factor: $dxdy = |J|dudv$).

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TikZ Schematic: Area Element Transformation under Jacobian Mapping



4. Distributions of Sums and Quotients

4.1 Distribution of the Sum $Z = X + Y$ (Convolution)

Derivation: PDF of the Sum of Two Continuous Random Variables

To find the PDF of $Z = X + Y$, introduce an auxiliary variable $W = X$. The forward transformation is:

$$z = x + y, \qquad w = x \tag{25}$$

The unique inverse transformation is:

$$x = w, \qquad y = z - w \tag{26}$$

Computing the Jacobian determinant:

$$J = \det \begin{pmatrix} \frac{\partial x}{\partial z} & \frac{\partial x}{\partial w} \\ \frac{\partial y}{\partial z} & \frac{\partial y}{\partial w} \end{pmatrix} = \det \begin{pmatrix} 0 & 1 \\ 1 & -1 \end{pmatrix} = (0)(-1) - (1)(1) = -1 \implies |J| = |-1| = 1 \tag{27}$$

The joint PDF of (Z, W) is:

$$f_{ZW}(z, w) = f_{XY}(w, z - w) \cdot |J| = f_{XY}(w, z - w) \tag{28}$$

Integrating out the auxiliary variable w yields the marginal PDF of the sum Z:

$$f_Z(z) = \int_{-\infty}^{\infty} f_{ZW}(z, w)\, dw = \int_{-\infty}^{\infty} f_{XY}(w, z - w)\, dw \tag{29}$$

If X and Y are **independent** ($f_{XY}(x, y) = f_X(x) f_Y(y)$), this becomes the fundamental **continuous convolution integral**:

$$f_Z(z) = (f_X * f_Y)(z) = \int_{-\infty}^{\infty} f_X(w) f_Y(z - w)\, dw = \int_{-\infty}^{\infty} f_X(z - y) f_Y(y)\, dy \tag{30}$$

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4.2 Distribution of the Quotient $Z = X/Y$

Derivation: PDF of the Quotient of Two Random Variables

Let $Z = \frac{X}{Y}$. Introduce the auxiliary variable $W = Y$. The forward transformation is:

$$z = \frac{x}{y}, \qquad w = y \quad (w \neq 0) \tag{31}$$

The inverse transformation is:

$$x = z \cdot w, \qquad y = w \tag{32}$$

The Jacobian determinant is:

$$J = \det \begin{pmatrix} \frac{\partial x}{\partial z} & \frac{\partial x}{\partial w} \\ \frac{\partial y}{\partial z} & \frac{\partial y}{\partial w} \end{pmatrix} = \det \begin{pmatrix} w & z \\ 0 & 1 \end{pmatrix} = (w)(1) - (z)(0) = w \implies |J| = |w| \tag{33}$$

The joint PDF of (Z, W) is:

$$f_{ZW}(z, w) = f_{XY}(zw, w) \cdot |w| \tag{34}$$

Integrating out w over $(-\infty, \infty)$ yields the general quotient density:

$$f_Z(z) = \int_{-\infty}^{\infty} |w| f_{XY}(zw, w)\, dw \tag{35}$$

If X and Y are independent:

$$f_Z(z) = \int_{-\infty}^{\infty} |w| f_X(zw) f_Y(w)\, dw \tag{36}$$

4.3 Quotient of Two Independent Standard Normals: The Cauchy Distribution

Classic Result: Ratio of Independent Standard Normals follows Cauchy

Let $X \sim \mathcal{N}(0, 1)$ and $Y \sim \mathcal{N}(0, 1)$ be independent. Their joint density is:

$$f_{XY}(x, y) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2} \cdot \frac{1}{\sqrt{2\pi}} e^{-y^2/2} = \frac{1}{2\pi} e^{-(x^2+y^2)/2} \tag{37}$$

Applying the quotient formula to $Z = X/Y$:

$$f_Z(z) = \int_{-\infty}^{\infty} |w| \cdot \frac{1}{2\pi} e^{-(z^2 w^2 + w^2)/2}\, dw = \frac{1}{2\pi} \int_{-\infty}^{\infty} |w| e^{-\frac{w^2(1+z^2)}{2}}\, dw \tag{38}$$

Since the integrand is even with respect to w:

$$f_Z(z) = \frac{2}{2\pi} \int_0^{\infty} w e^{-\frac{w^2(1+z^2)}{2}}\, dw = \frac{1}{\pi} \int_0^{\infty} w e^{-\frac{w^2(1+z^2)}{2}}\, dw \tag{39}$$

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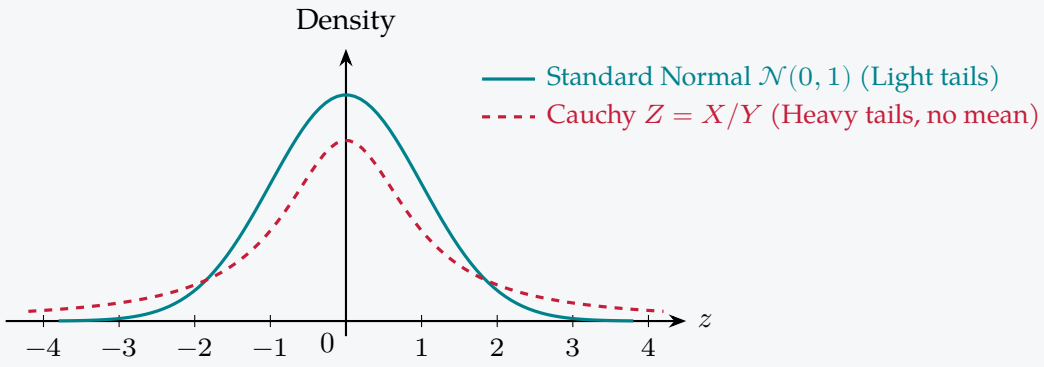
Substitute $u = \frac{w^2(1+z^2)}{2} \implies du = (1+z^2)wdw \implies wdw = \frac{du}{1+z^2}$:

$$f_Z(z) = \frac{1}{\pi(1+z^2)} \int_0^\infty e^{-u}\, du = \frac{1}{\pi(1+z^2)} \left[-e^{-u}\right]_0^\infty = \frac{1}{\pi(1+z^2)}(1) \tag{40}$$

$$f_Z(z) = \frac{1}{\pi(1+z^2)}, \qquad -\infty < z < \infty \tag{41}$$

This is the celebrated **Standard Cauchy Distribution**! *Crucial Exam Note:* For a Cauchy distribution, $\int_{-\infty}^{\infty} |z| \frac{1}{\pi(1+z^2)} dz = \infty$. Therefore, the Cauchy distribution has **no defined mean, no variance, and no MGF**!

TikZ Plot: Standard Cauchy PDF vs. Standard Normal Gaussian PDF



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Quick Revision — Module 3

Bivariate Distributions, Sums, Quotients and Bayes' Rule | BS-M301

Key Formulas and Bivariate Relations at a Glance

Joint PDF & CDF	$f_{XY}(x,y) \geq 0,\ \iint f_{XY}dxdy = 1; \quad f_{XY}(x,y) = \frac{\partial^2 F_{XY}}{\partial x \partial y}$
Marginal Densities	$f_X(x) = \int_{-\infty}^{\infty} f_{XY}(x,y)dy, \quad f_Y(y) = \int_{-\infty}^{\infty} f_{XY}(x,y)dx$
Conditional Densities	$f_{X\|Y}(x \mid y) = \frac{f_{XY}(x,y)}{f_Y(y)}, \quad f_{Y\|X}(y \mid x) = \frac{f_{XY}(x,y)}{f_X(x)}$
Statistical Independence	$f_{XY}(x,y) = f_X(x)f_Y(y) \iff f_{X\|Y}(x \mid y) = f_X(x)$
Tower Property	$E\left[E[X \mid Y]\right] = E[X], \quad \text{Var}(X) = E[\text{Var}(X\|Y)] + \text{Var}(E[X\|Y])$
Jacobian Transform	$f_{UV}(u,v) = f_{XY}(h_1,h_2) \cdot \lvert J\rvert, \quad J = \det \begin{pmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{pmatrix}$
Sum & Quotient PDFs	$Z = X + Y \implies f_Z = \int f(w, z-w)dw; \quad Z = \frac{X}{Y} \implies f_Z = \int \lvert w\rvert f(zw,w)dw$
Cauchy & Bayes' Rule	$Z = \frac{X}{Y} \sim \text{Cauchy} : f_Z(z) = \frac{1}{\pi(1+z^2)}; \quad f_{X\|Y}(x \mid y) = \frac{f_X(x)f_{Y\|X}(y\|x)}{\int_{-\infty}^{\infty} f_X(u)f_{Y\|X}(y\|u)du}$

One-Line Definitions

- **Joint PDF $f_{XY}(x,y)$:** Surface whose double integral over any region D equals $P((X,Y) \in D)$.
- **Marginal PDF:** Individual density of one component obtained by integrating out the other variable.
- **Conditional PDF:** Sliced cross-sectional density of X normalized by $f_Y(y)$ under $Y = y$.
- **Statistical Independence:** Factorization $f_{XY}(x,y) = f_X(x)f_Y(y)$ across the entire domain.
- **Tower Property:** Fundamental law asserting that overall expectation equals $E[E[X \mid Y]]$.
- **Jacobian Determinant (J):** Local area transformation ratio $dxdy = \lvert J\rvert dudv$ relating coordinate spaces.
- **Convolution Theorem:** Integral operation yielding the PDF of the sum of independent RVs: $f_Z = f_X * f_Y$.
- **Quotient Distribution:** Density $f_Z(z) = \int \lvert w\rvert f(zw,w)dw$ describing the ratio of two RVs.
- **Cauchy Distribution:** Heavy-tailed distribution $f(z) = \frac{1}{\pi(1+z^2)}$ formed by $\mathcal{N}(0,1)/\mathcal{N}(0,1)$, lacking moments.
- **Posterior Density:** Updated density of an unknown parameter after incorporating observation y.

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Frequently Asked / Viva Questions (Part 1)

Q1. State the necessary and sufficient conditions for two continuous random variables X and Y to be independent.

X and Y are independent if and only if their joint PDF factors into the product of their marginal PDFs everywhere: $f_{XY}(x,y) = f_X(x)f_Y(y)$ for all $(x,y) \in \mathbb{R}^2$, and the support region D is a product of intervals (non-coupled boundaries).

Q2. How is the joint PDF obtained from the joint Cumulative Distribution Function?

By mixed second-order partial differentiation: $f_{XY}(x,y) = \frac{\partial^2}{\partial x \partial y} F_{XY}(x,y)$, provided the second partial derivatives exist and are continuous.

Q3. State and prove the Law of Iterated Expectations (Tower Property).

$E\left[E[X \mid Y]\right] = E[X]$. Proof: $E[E[X \mid Y]] = \int \left(\int x \frac{f(x,y)}{f_Y(y)} dx \right) f_Y(y) dy = \iint x f(x,y) dx dy = \int x f_X(x) dx = E[X]$.

Q4. What is the physical significance of the Jacobian determinant in bivariate transformations?

The Jacobian determinant $J = \frac{\partial(x,y)}{\partial(u,v)}$ represents the local geometric dilation factor between the original and transformed coordinate planes, ensuring conservation of probability mass: $P(dxdy) = f_{XY}(x,y)dxdy = f_{XY}(h_1,h_2)|J|dudv = f_{UV}(u,v)dudv$.

Q5. Derive the PDF of the sum $Z = X + Y$ of two independent continuous random variables.

Let $Z = X + Y, W = X \implies X = W, Y = Z - W$. The Jacobian is $|J| = 1$. The joint density is $f_{ZW}(z,w) = f_{XY}(w, z-w) = f_X(w)f_Y(z-w)$. Integrating over w gives the convolution: $f_Z(z) = \int_{-\infty}^{\infty} f_X(w)f_Y(z-w)dw$.

Q6. Why does the absolute value $|w|$ appear in the formula for the quotient density $Z = X/Y$?

The Jacobian of the inverse transformation $x = zw, y = w$ is $J = \det \begin{pmatrix} w & z \\ 0 & 1 \end{pmatrix} = w$.

Because an area element must always be strictly non-negative ($dxdy = |J|dzdw$), the absolute value $|J| = |w|$ must be taken.

Q7. Show that the ratio of two independent standard normal random variables follows a Cauchy distribution.

$f_Z(z) = \frac{1}{2\pi} \int_{-\infty}^{\infty} |w| e^{-w^2(1+z^2)/2} dw = \frac{1}{\pi} \int_0^{\infty} w e^{-w^2(1+z^2)/2} dw$. Substituting $u = w^2(1+z^2)/2$ transforms the integral to $\frac{1}{\pi(1+z^2)} \int_0^{\infty} e^{-u} du = \frac{1}{\pi(1+z^2)}$, which is the standard Cauchy PDF.

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Frequently Asked / Viva Questions (Part 2)

Q8. Why does the Cauchy distribution have no mathematical expectation or variance?

$E[|Z|] = \frac{1}{\pi}\int_{-\infty}^{\infty}\frac{|z|}{1+z^2}dz = \frac{2}{\pi}\int_0^{\infty}\frac{z}{1+z^2}dz = \frac{1}{\pi}\left[\ln(1+z^2)\right]_0^{\infty} = \infty$. Because the absolute integral diverges logarithmically due to heavy polynomial tails ($\sim 1/z^2$), the mean is mathematically undefined, and all higher moments diverge.

Q9. State the Law of Total Variance (Eve's Law) and interpret its terms.

$\text{Var}(X) = E[\text{Var}(X \mid Y)] + \text{Var}(E[X \mid Y])$. The first term represents the average unexplained variation remaining in X after conditioning on Y; the second term represents the variation in the conditional predictor $E[X \mid Y]$ explained by Y.

Q10. State Continuous Bayes' Theorem and explain its role in signal estimation.

$f_{X|Y}(x \mid y) = \frac{f_X(x)f_{Y|X}(y|x)}{\int f_X(u)f_{Y|X}(y|u)du}$. In communications, X is the transmitted signal (prior), $f_{Y|X}(y \mid x)$ is the channel noise model (likelihood), and $f_{X|Y}(x \mid y)$ is the posterior distribution used by the receiver to make Maximum A Posteriori (MAP) symbol decisions.

Summary Matrix: Comprehensive Comparison of Bivariate Operations

Operation / Concept	Mathematical Governing Expression	Independence Simplification	Physical Meaning / Role
Marginal Density	$f_X(x) = \int_{-\infty}^{\infty} f_{XY}(x,y)\,dy$	$f_X(x)\int f_Y(y)dy = f_X(x)$	Complete projection onto X-axis.
Conditional Density	$f_{X\|Y}(x \mid y) = \frac{f_{XY}(x,y)}{f_Y(y)}$	$f_{X\|Y}(x \mid y) = f_X(x)$	Updated density given knowledge $Y = y$.
Sum of RVs	$Z = X + Y \implies f_Z(z) = \int f_{XY}(w, z-w)dw$	Convolution: $f_Z = f_X * f_Y$	Total noise from independent sources.
Quotient of RVs	$Z = X/Y \implies f_Z(z) = \int \lvert w\rvert f_{XY}(zw, w)dw$	$\int \lvert w\rvert f_X(zw)f_Y(w)dw$	Signal-to-noise ratio; slope of vector.
Cauchy Ratio	$Z = X/Y$ for $X, Y \sim \mathcal{N}(0,1) \implies \frac{1}{\pi(1+z^2)}$	Quotient of two standard Gaussians	Heavy-tailed ratio; undefined moments.
Continuous Bayes	$f_{X\|Y}(x \mid y) = \frac{f_X(x)f_{Y\|X}(y\|x)}{\int f_X(u)f_{Y\|X}(y\|u)du}$	$f_{X\|Y}(x \mid y) = f_X(x)$	Optimal Bayesian estimation / inference.

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